

0, 1, 2, 3, 4, 5, 6
X

Lec 7

Practice Questions on time
Complexity

$a=0, b=0$

(16)

$a = 0, 1, 2, 3, 4$

$b = 0$

17.) main() →
{

$a=0, b=0;$ →

→ for(int i=1; i<=n; i=i*2)

{ $a++;$ } $a \leftarrow a+1$

→ for(int j=1; j<=a; j=j*2)

{ $b++;$ }

1.) Value of a, b?

2.) TC?

Time complexity?

$n = \log_2 n$

$a = \log_2 n$

int i=1;

i<=n,
n=100

$i = i * 2$

$i = 1$

$i <= 100$

$i = i * 2$

$i = 2$

$i <= 100$

$i = 2 * 2$

$i = 4$

$i <= 100$

$i = 4 * 2$

$i = 8$

1, 2, 4, 8
 $2^0, 2^1, 2^2, 2^3$

$i = 2^x$

$i \geq 100$

$2^x \geq n$

$x = \text{how many times loop is over}$

for (i=1; i <= n; i = i+1)

$$2^x = n$$

$$x = \log_2 n$$

for (j=1; j <= log n; j = j*2)

$$\Rightarrow \log_2 2 = \log_2 n$$

{ b++;

} j = 2^x

j >= log n

$$2^x \geq \log_2 n$$

$$\Rightarrow x \log_2 2 \geq \log_2 (\log_2 n)$$

$$\Rightarrow x = \log_2 \log_2 n$$

n = how many times loop executed

$$b = \log_2 \log_2 n$$

$\Rightarrow$ Time complexity $\Rightarrow$ how many times loop is getting executed

$$\text{Loop 1} \Rightarrow \log_2 n$$

$$\text{Loop 2} \Rightarrow \log_2 \log_2 n$$

$$TC = \log_2 n + \log_2 \log_2 n$$

$$\checkmark O(\log n + \log \log n) \Rightarrow O(\log n)$$

$$\boxed{O(n^2 + n + 1) = O(n^2)}$$

$$\boxed{O(n^{25} + n^3 + n^6) = O(n^{25})}$$

$\Theta, O, \omega, \Omega$

$$\frac{a=18}{26}$$

$$a=0, \quad b=0$$

18.) main() $a=0, b=0;$

```

{ for(int k=1; k<=n; k++)
{ a=0; b=0;
  for(int i=1; i<=n; i=i*2)
  { a++;
    for(int j=1; j<=a; j=j*2)
    { b++;
    }
  }
}
}

```

$$k=1, \quad a=\log n, \quad b=\log \log n$$

$$k=2, \quad a=2 \log n, \quad b=2 \log \log n$$

$$k=n, \quad a=n \log n, \quad b=n \log \log n$$

$$a = n \log n$$

$$b = n \log \log n$$

$$T_c = n \log n + n \log \log n$$

$$= n (\log + \log \log n)$$

$$= n \log n$$

$$T_c = O(n \log n)$$

1.) Value of a, b?

2.) TC?

19.) main()

```

{ for(int k=1;k<=n;k=2*k)
{ a=0,b=0; a=0; b=0;
  for(int i=1;i<=n^2;i=i*7)
  { a++;
    for(int j=n^3;j>=7;j=j/9)
    { b++;
    }
  }
}

```

1.) Value of a,b?

2.) TC?

$$j < 7$$

$$\frac{n^3}{9^u} = 7$$

$$k=2$$

$$j = n^3 / 9^0 \quad j \geq 7$$

$$j = n^3 / 9 \quad n^3 / 9 \geq 7$$

$$j = n^3 / 9^2$$

$$j = \frac{n^3}{9^u}$$

$$j = j / 9$$

$$j = (n^3 / 9) / 9$$

$$a = 2 \log_7 n$$

$$b = 3 \log_9 n - \log_7 7$$

$$k=1$$

$$a=0$$

$$b=0$$

$$i=1$$

$$i \leq n^2$$

$$7 \leq n^2$$

$$7^2 \leq n^2$$

$$7^k = n^2$$

$$n \log_7 7 = 2 \log_7 n$$

$$n = 2 \log_7 n$$

$$n = 2 \log_7 n$$

$$a = 2 \log_7 n$$

$$b = 3 \log_9 n - \log_7 7$$

$$k=1, \quad k \leq n, \quad k=2^*k$$

$$k=2$$

$$k=4$$

$$2^n = n$$

$$n = \log_2 n$$

$$\log n$$

$$\frac{n^3}{9^n} = 7$$

$$\Rightarrow n^3/7 = 9^n$$

$$\Rightarrow \log_9(n^3/7) = \frac{2 \log_9 9}{1}$$

$$\Rightarrow n = \log_9 n^3 - \log_9 7$$

$$n = 3 \log_9 n - \log_9 7$$

$$\log(a/b) = \log a - \log b$$

$$b = 3 \log_9 n - \log_9 7$$

$$\log_a b = \frac{\log_m b}{\log_m a}$$

Time complexity = how many times loop is executed.

$$T(n) = (a + b) \log_2 n$$

$$= (2 \log_7 n + 3 \log_9 n - \log_9 7) \log_2 n$$

$$\Rightarrow (2 \log_7 n \cdot \log_2 n + 3 \log_9 n \times \log_2 n)$$

$$O((\log_2 n)^2) + 3(\log_2 n)^2$$

$$O((\log_2 n)^2)$$

$$O((\log_2 n)^2)$$

$$\log_7 n$$

$$\frac{\log_2 n}{\log_2 7}$$

Thamir

